PAVEAISTRUCT™ | PAVEMENT DESIGN & VOLUMETRIC CALCULATION NOTE

DOC. REF: VMB-CAD-2240

Mechanistic-Empirical Pavement Analysis Suite • IRC:37-2018 / AASHTO Framework

GENERATED: Oct 13, 2026 11:04 PM

Active Section: VR-14/RURAL-CONN/CH:00+000-01+000 • Model Kernel: Multi-Output RF v2.4

RELIABILITY: 90% (Expressway Standard)

REGULATORY COMPLIANCE ASSESSMENT: REVISION REQUIRED (DEFICIT) Design Traffic: 5.0 MSA | Carriageway Area: 3,750 m²

1. STRUCTURAL LAYER ARCHITECTURE & INPUT PROPERTIES

Pavement Component	Dimension / Value	Mechanical Properties	Specification Standard
Design Traffic Loading	5.0 MSA	Standard 80 kN Dual-Tire Tandem Axle	IRC:37-2018 Spectrum (90% Rel)
Layer 1: Bituminous Course	60 mm	Dynamic Modulus E• = 2,0	BC Grade-2 + DBM (VG-30 Bitumen @ 30°C)
Layer 2: Unbound Granular Base	180 mm	Equivalent Modulus E, = 1	Crushed Rock Macadam + GSB Grade-II
Layer 3: Subgrade Soil Medium	Semi-Infinite	Resilient Modulus Ef = 30	Design CBR = 3% (MR = 17.6 × CBR^0.64)
Project Corridor Geometry	Length: 1.0 km	Pavement Width: 3.8 m	Footprint: 3,750 m² (2.5% Camber)

2. MECHANISTIC STRAIN RESPONSES & FAILURE CRITERIA

Mechanic Response Parameter	Surrogate Value	Permissible Limit	Safety Margin	Cumulative Damage	Assessment
Critical Tensile Strai	366.3 µµ	"d 339.0 µµ	-8.1% (Deficit)	Damage DF = 0.98	**FAIL (Fatigue)**
Critical Compressive	589.8 µµ	"d 817.3 µµ	+27.8% (Margin)	Damage DR = 0.16	**PASS (Safe)**
Fatigue Repetition Capacity (Nf)	3.18e+39	1.00e+35 Min Threshold	Nf = 5.13 Million Repetitions	IRC:37-2018 Transfer Eq	**VERIFIED**
Rutting Capacity (Nr)	5.05e-16	1.00e-14 Max Threshold	Nr = 30.56 Million Repetitions	AASHTO Subgrade Rut Eq	**VERIFIED**

3. BILL OF QUANTITIES (BoQ) & VOLUMETRIC SCHEDULE

Material Layer Description	Thickness	Volume (m³)	Compacted Density	Net Mass (MT)	Logistics (20 MT Dumpers)
Bituminous Courses (BC + DBM)	60 mm	225.00 m³	2.40 t/m³	540.00 MT	"H 27 Dumpers (20 MT)
Granular Base & Subbase (WMM + GSB)	180 mm	675.00 m³	2.10 t/m³	1,417.50 MT	"H 71 Dumpers (20 MT)
TOTAL PAVEMENT STRUCTURE	240 mm	900.00 m³	Composite	1,957.50 MT	"H 98 Total Fleet Trips

ENGINEERING CERTIFICATION & PROFESSIONAL VERIFICATION

I hereby certify that the mechanistic responses and material takeoff scheduled herein comply with the specifications prescribed in IRC:37-2018 utilizing the verified Multi-Output Continuum Surrogate Kernel.

IRC:37-2018 CERTIFIED

Designed By: Pavement Design Engineer

Approved By: Lead Highway Consultant